



Project 2 – Pick and Place Machine

CPEN307 – PLC Lab

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Introduction:

This project focuses on the development and programming of an automated pick-and-place loading system controlled by a PLC. The goal is to reliably transfer sheets from a stacked supply area onto a conveyor using coordinated horizontal and vertical movements, sensor feedback, and vacuum gripping. We designed the control logic to ensure safe operation, accurate sheet handling, and continuous cycling as long as material is available, and the conveyor remains clear.

Description:

The system is an automatic pick-and-place loader designed to transfer sheets from a stacked supply area onto a conveyor. The process begins when the operator presses the START button, enabling automatic mode. Before any movement starts, we programmed the PLC to verify two conditions: the Stack_Sensor must detect available sheets in the stack, and the Conveyor_Sensor must confirm that the conveyor is empty and ready to receive a new sheet.

Once these conditions are satisfied, the horizontal axis moves toward the stack. This movement continues until the HorizontalSensorBack becomes active, indicating correct positioning over the stack. The vertical axis then lowers the gripper until the VerticalSensorDownS detects that the suction cup has reached the sheet. At this point, the PLC activates the vacuum valve, allowing the gripper to secure the top sheet. After a brief confirmation delay, the vertical axis lifts the sheet back to its upper position.

With the sheet lifted, the horizontal axis travels toward the conveyor side until the HorizontalSensorForward becomes active. The vertical axis then lowers the sheet onto the conveyor, stopping when VerticalSensorDownC is triggered. The PLC deactivates the vacuum valve, releasing the sheet onto the conveyor surface. The vertical axis lifts again, and the horizontal axis returns to the stack position to begin the next cycle, as long as new sheets are available and the conveyor remains clear.

Each time a complete pick-and-place cycle is finished, the PLC sets a finish signal. We use a rising-edge detection circuit to count each cycle only once. The system tracks the number of

completed cycles, and once the counter reaches five cycles, a comparator triggers a reset, returning the counter to zero and allowing a new batch count to begin.

Pressing the STOP button at any time brings the machine to a controlled halt, safely stopping all motion.

I/O Table (Inputs, Outputs, Memory Bits):

Digital Inputs (I):

Symbol	Address	Type	Description
Conveyer_Sensor	I124.7	BOOL	Detects sheet on conveyer
HorizontalSensorBack	I124.6	BOOL	Gantry at stack position
HorizontalSensorForward	I124.7	BOOL	Gantry at conveyer position
Stack_Sensor	I124.4	BOOL	Detects sheet presence in stack
Start	I124.0	BOOL	START push button
Stop	I124.1	BOOL	STOP push button (Normally closed)
VerticalSensorDownC	I125.1	BOOL	Vertical axis – down at conveyer
VerticalSensorDownS	I125.0	BOOL	Vertical axis – down at stack

Digital Outputs (Q):

Symbol	Address	Type	Description
Conveyer	Q124.1	BOOL	Conveyor motor output
Cylinder_Back	Q124.3	BOOL	Horizontal axis – move back
Cylinder_Down	Q124.5	BOOL	Vertical axis – move down
Cylinder_Forward	Q124.6	BOOL	Horizontal axis – move forward
Cylinder_Up	Q124.6	BOOL	Vertical axis – move up

Lamp	Q124.0	BOOL	Status lamp
Stop	Q124.4	BOOL	STOP output (if used to reset circuits)
Vacum	Q124.3	BOOL	Vacuum valve output

Memory Bits (M):

Symbol	Address	Type	Description
StartM	M124.0	BOOL	Internal start latch
Vacuum_release	M124.1	BOOL	Vacuum release control
Vacuum_On	M124.2	BOOL	Vacuum enabled state
StartAgain	M124.3	BOOL	Restart cycle memory bit
Finish	M124.4	BOOL	Cycle finished flag

System Layout Diagram

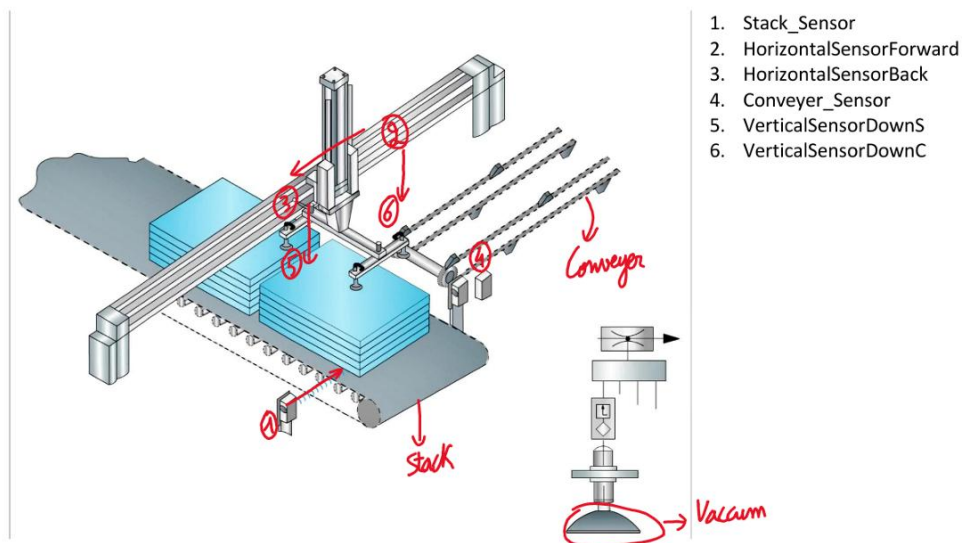


Figure: Assigned sensor labels and their physical locations within the pick-and-place mechanism.